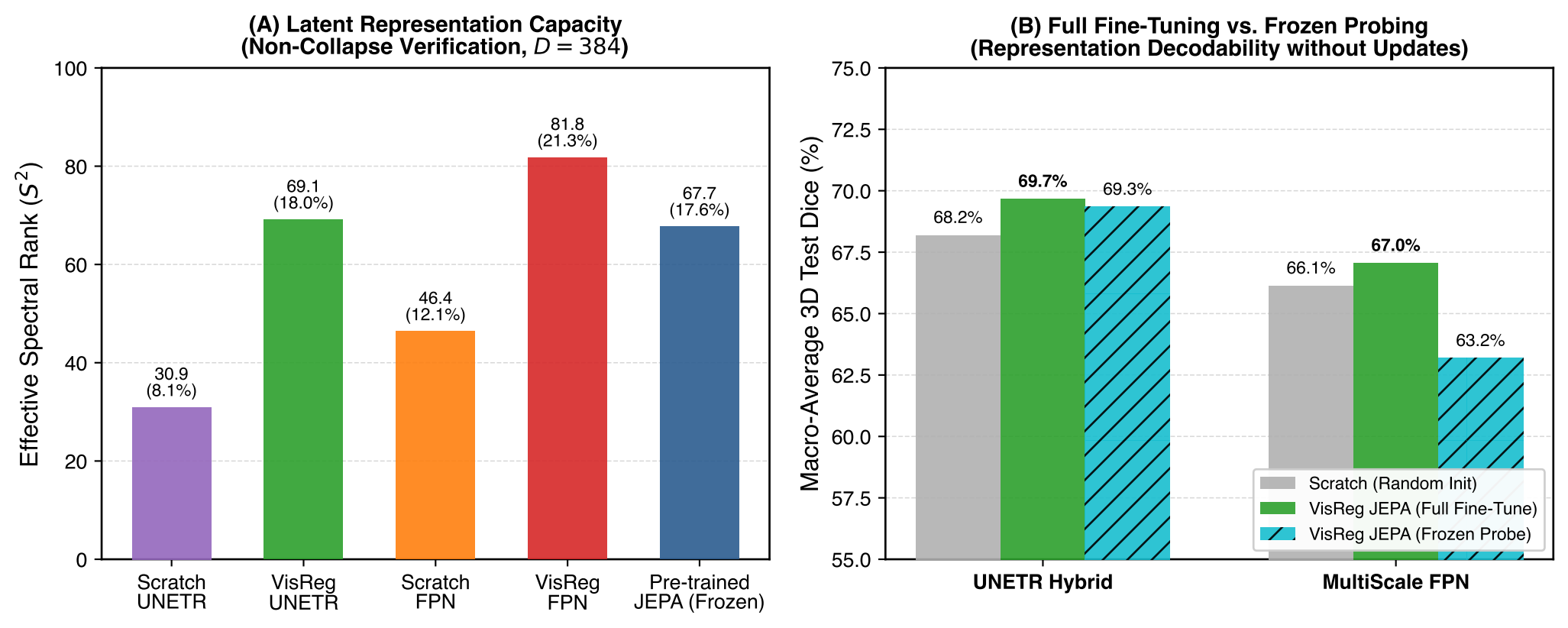
(A) Latent Representation Capacity
(Non-Collapse Verification, $D = 384$)
Effective Spectral Rank (S^2)
100
80
60
40
20
0
30.9
(8.1%)
69.1
(18.0%)
46.4
(12.1%)
81.8
(21.3%)
67.7
(17.6%)
Scratch
UNETR
VisReg
UNETR
Scratch
FPN
VisReg
FPN
Pre-trained
JEPA (Frozen)
(B) Full Fine-Tuning vs. Frozen Probing
(Representation Decodability without Updates)
Macro-Average 3D Test Dice (%)
75.0
72.5
70.0
67.5
65.0
62.5
60.0
57.5
55.0
68.2%
69.7%
69.3%
66.1%
67.0%
63.2%
UNETR Hybrid
MultiScale FPN
Scratch (Random Init)
VisReg JEPA (Full Fine-Tune)
VisReg JEPA (Frozen Probe)